

* Father of genetics

Theories before Mendel began his experimentations

① Pangenesis

- * particles known as **gemmules**, carried out information from various parts of the body to the reproductive organs so that they can be passed on to the embryo

② Preformationism

- * inside the egg or sperm, there exists a mini human known as **homunculus**, which only enlarges during development
- * This meant that all traits would be inherited from only one parent

③ Inheritance of acquired characteristics

- * The acquired during one's lifetime will become incorporated into one's hereditary information, which is later passed down

④ Blending theory of inheritance

- * Both sexes contribute equally to the new individual "offspring is a blend/mixture of parents"
- * Parents of contrasting appearance, will always produce an offspring with that appearance
- * Ex) red flower + white flower $\xrightarrow{\text{always}}$ pink flower (but red & white reappear in future generations)
- * Once Blended, the genetic differences can't be separated

⑤ Germ-plasm theory "Chromosomal theory"

- * The reproductive cells carry the complete set of genetic information to gametes which is later passed down to offspring

homozygous		T	T
		Tt	Tt
Recessive (test cross)	t	Tt	Tt
	t	Tt	Tt

heterozygous		T	t
		Tt	tt
Recessive (test cross)	t	Tt	tt
	t	Tt	tt

Mendel's Laws

- * Mendel discovered the basic principles of hereditary by breeding pea plants

- * Because pea plants are cheap - short lifecycle - counter traits "opposing traits"

Law of Segregation	Alleles can segregate and recombine
Law of Independent Assortment	Traits can segregate and recombine independently of other traits

① Mendel's first Law The principle of segregation

- * Every individual carries a pair of factors for each trait and the members of the pair of alternative characters are separated during the formation of gametes

Assessment

Genetics of Cells and Traits

Tall

Monohybrid cross

T T

short

t

T	T	T
t	T	T

offspring: gen 1 Tt (tall) 100%

gen 1 (F1) will always have the same phenotype

gen 1

Offspring

T t

TT

tall

25%

genotype

Phenotype

gen 1

T

TT

Tt

Tt

tall

50%

ratio

1:2:1

3:1

t

Tt

tt

tt

short

25%

genotype

Phenotype

② Mendel's second law the principle of independent assortment

- * When gametes are formed, alleles of a gene for one trait segregate independently from the alleles of a gene of a different trait

		Dihybrid cross					
		TG	Tg	tG	tg	9:3:3:1	
T: tall	TG	TTGG	TtGg	TtGg	TtGg	9 tall - green	
t: short	Tg	TTGg	TtGg	TtGg	TtGg	3 tall - yellow	
G: green	tG	TtGg	TtGg	TtGg	TtGg	3 short - green	
g: yellow	tg	TtGg	TtGg	TtGg	TtGg	1 short - yellow	

period when it is potentially hermaphroditic(bipotential gonads). By the 5th week of gestation, gonadal primordia (the tissue that will form the gonad) arise as a pair of ridges associated with each embryonic kidney.

Primordial germ cells migrate to these ridges, Where an outer cortex and inner medulla formed. The cortex is capable of developing into an ovary, while the inner medulla may develop into a testis.

If the cells of the genital ridge have the XY constitution (in presence of SRY gene), development of the medullary region into a testis is initiated around 7th week. However, in absence of the Y chromosome, no male development occurs, and the cortex of the genital ridge subsequently forms ovarian tissue under the actions of estrogen hormone.

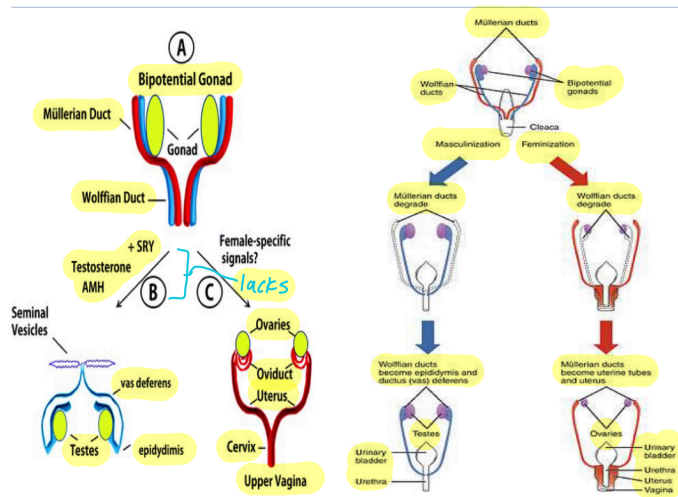
female(müllerian) duct systems degenerates.

Once male differentiation is initiated, the embryonic testicular tissue secretes two hormones (Testosterone and anti müllerian hormones) that are essential for continued male sexual differentiation.

- Testosterone hormone secreted by Leydig cells which stimulates the development of epididymis, vas deferens, seminal vesicle, and ejaculatory duct.

- Anti müllerian hormone secreted by Sertoli cells to degenerates the female müllerian ducts

Testosterone converted to a potent androgen DHT (dihydro testosterone) by 5 α reductase enzyme. DHT induces masculinization and development of urethra, prostate and external genitalia (penis and scrotum)



Sex-linked inheritance "X-linked or Y-linked"

* Specific genes located on either x or y chromosome

Visible light

Violet 400nm → Red 700nm

X-linked recessive inheritance

① color blindness

- * 3 genes are involved in distinguishing color & there are 3 types of cone cells in the eye
- * The genes coding for the pigments Red & green are found on the X chromosome
- * The gene coding for the pigment Blue is found on autosome no. 7
- * The eyes are 95% rods cells for light & 5% cone cells for color

In males

- * Defect in the gene for green → cant distinguish between red & green
- * Defect in the gene for red → red will appear as green

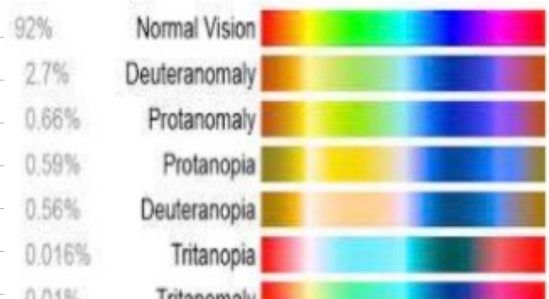
X^cX^c Female - normal vision

X^cX^C Female - normal vision, but is a carrier for color blindness

X^cX^c Female - Color blind

X^CY Male - normal vision

X^cY Male - Color blind



① Monochromacy

- one or no type of cone cells are present
- known as rod monochromacy or achromatopsia
- true color blindness, the can only see shades of grey ranging from Black to white

② Dichromatic

- is caused by the missing of one of the three retinal cones

① Protanopia

- absence of red cones
- difficulties in distinguishing between red & green

② Deuteranopia

- absence of green cones
- some grey-purple-greenish blue can not be distinguished clearly

③ Tritanopia

- absence of blue cones
- Blue appears greenish-yellow & oranges appear pinkish

③ Anomalous Trichromacy

- mild or weak color vision deficiency
- is when all 3 cones are present, but one of them has shifted spectral sensitivity
- difficult to distinguish between shades of different colors

① Protanomaly weak red cones

② Deuteranomaly weak green cones

③ Tritanomaly weak blue cones

Causes of color vision deficiency

- ① acquired with aging
- ② Diabetes
- ③ Malignancy / tumor in the eye
- ④ physical or chemical injury
- ⑤ Radiation "UV"
- ⑥ Ingestion of drugs / chemicals

② Hemophilia

- Hemophilia A is most common severe hereditary disease
- Hemophiliacs bleed severely when injured, because they lack factor VIII which is a plasma protein and is the key component for clotting
- Fibrin fails to be produced therefore severe prolonged bleeding will result even from minor injuries
- Person with hemophilia A can be treated with clotting factor VIII extracted from a normal person's blood

Hereditary bleeding disorders

① Hemophilia deficiency of factor VIII

- severe patient produces 1% of factor VIII - experience frequent spontaneous bleeding several times per month
- moderate patient produces 5% of factor VIII - experience frequent bleeding several times per year after mild trauma
- mild patient produces 25-45% of factor VIII - experience bleeding after surgery or severe trauma

② Hemophilia missing or deficiency of factor IX

③ Hemophilia deficiency of factor XI

④ Parahemophilia deficiency of factor V

⑤ von Willebrand disease missing or defective factor VWF

Duchenne muscular dystrophy / degenerating muscles / loss of coordination

- Signs & symptoms appear before the age of 5 in early childhood & death occurs usually before age 20
- All muscles will eventually degenerate therefore leading to cardiac & respiratory failure
- This is due to the lack of the protein **dystrophin** coded for by the DMD gene

④ Ichthyosis "fish scales disease"

- rough brown scaly skin
- enzyme deficiency of **steroid sulfatase** which blocks the removal of cholesterol from skin cells therefore the upper skin layer won't shed off therefore will accumulate and stick to itself

X-linked Dominant Inheritance

- Less common than X-linked recessive

Ex) Hyperphosphatemic Rickets

- decreased ability of the kidney to reabsorb phosphate leading to ossification / bone softness / bending

① The Fragile X Syndrome "Martin-Bell Syndrome"

- even though it's dominant and not recessive, females are still less likely to get it due to **X-inactivation in females**
- Characterized by large ears - long face - protruding chin - hyper mobile joints - macroorchidism in post pubertal males
- mild mental retardation that's more variable in females than males

② Congenital generalized hypertrichosis "hair over growth that covers the body and is present at birth"

- produces more hair follicles leading to denser & more upper body hair
- hair growth is milder & patchier in females due to hormonal differences

Y-linked Inheritance "holandric inheritance"

- Y chromosome contains fewer genes than X
- genes control differentiation of embryo into male & encoding testis-specific spermatogenesis factors
- transmission of disease is strictly father - to - son

Sex limited & Sex-Influenced Traits

- Sex limited trait occurs in only one sex due to anatomical differences
- Ex) inherited uterine defects in females - inherited testicular defects in male
- most common example is male pattern baldness which is more common in males
- it's believed that in males it is inherited autosomal dominant trait, but in females inherited autosomal recessive trait
- usually homozygous females express pattern baldness, but heterozygous females with adrenal tumors also express pattern baldness due to high testosterone

Thalassemia

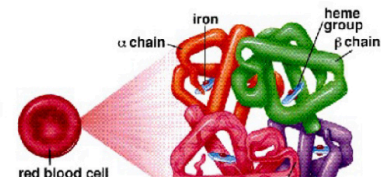
- **Hemoglobinopathies** are mutations which cause a change in the primary structure of one of the globin chains
- **Thalassemias** are mutations which alter the level of expression of one of the globin chains
- They are heritable, hypochromic anemias which result from decreased or absence of globin chains synthesis
- more common in Asia - Africa - Middle east

Hemoglobin

- in adults Hgb A₁ major Hgb A₂ minor Hgb F fetal
- protein that consists of
 - 4 subunits "polypeptide chains that aren't covalently bonded"
 - 4 heme complexes with Iron bound "1 heme group per subunit"

Hemoglobin Structure

- Four subunits
 - two α
 - two β
- Iron
- Heme
- Binds 4 O₂



Alpha globin

- 141 amino acids
- coded for on chromosome 16
- 2 genes on each chromosome
- 4 alleles in each diploid cell
- Gene deletion will result in Alpha-thalassemias
- found in Hgb A₁-A₂-F and Gower 2

Beta globin

- 146 amino acids
- coded for on chromosome 11
- 1 genes on each chromosome
- 2 alleles in each diploid cell
- Point mutations will result in Beta-thalassemias
- found in Hgb A₁ and portland 2

Delta globin

- coded for on chromosome 11
- found in Hgb A₂ - small amounts in all adults

Gamma globin

- coded for on chromosome 11
- found in Hgb F and portland 1

Zeta

- coded for on chromosome 16
- found in embryonic hemoglobin "Gower 1 and portland 1 & 2"

Epsilon

- coded for on chromosome 11
- found in Gower 1 and Gower 2

Hemoglobin types

Hgb A₁ major

Hgb A₂ minor

Hgb F fetal

Gower 1

Gower 2

Portland 1

Portland 2

Hgb H

Bart's Hgb

Hgb S sickle cell anemia

Hgb C

Globin chains / subunits

2 Alpha α 2 Beta β adult

2 Alpha 2 Delta δ adult

2 Alpha 2 Gamma γ adult & fetal

2 Zeta ζ 2 Epsilon ε embryo

2 Alpha 2 Epsilon embryo

2 Gamma 2 Zeta embryo

2 Beta 2 Zeta embryo

4 Beta

4 Gamma

2 Alpha 2 Beta ^{location} 6 glu → val

2 Alpha 2 Beta ^{location} 6 glu → lys

Types of Point Mutations

① frameshift

TAT TGG CTA GTA CAT

Tyr Trp Leu Val His

TAC TCG GCT AGT ACA T

Tyr Ser Ala Ser Thr

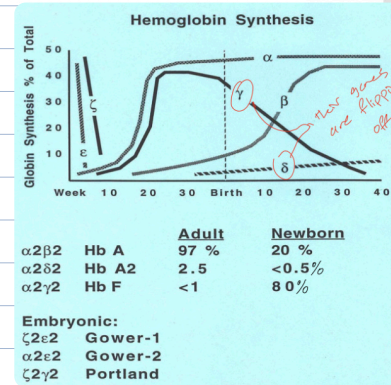
DNA level

mRNA level

protein level

② Silent ③ Nonsense ④ Missense

DNA level	TTT	ATC	TCC	TGC
mRNA level	AAA	UAG	AGG	ACG
protein level	Lys	STOP	Arg	Thr



All globin chains are present in a human, but certain ones tend to switch off or on during the transition from embryo to adult

Genetic basis

Polygenic trait is when one's phenotype is influenced by more than one gene

Pleiotropic gene is a single gene that controls more than one trait

missense mutation "Base change" is where genetic material has coded for an amino acid but not the desired one

nonsense mutation stop codon is coded before the full protein synthesis, leading to termination

Silent mutation there is a mistake in the genetic material, but doesn't change the resulting amino acid

frameshift mutation caused by an addition or deletion of a base pair resulting in translation of the genetic material, meaning there will be a shift in the whole sequence

Classification of Alpha Thalassemia

- ① one deletion is a silent carrier with no clinical significance
- ② two deletions is an α Thal. minor with mild hypochromic microcytic anemia
- ③ three deletions is Hgb H has variable severity, but is less severe than β Thal major
- ④ four deletions is Bart's Hgb hydrops fetalis, in utero or early neonatal death

the absence of alpha chains will result in an increase & excess of gamma globin chains during fetal life & an excess of Beta globin chains later in postnatal life

Beta Thalassemia

- results from point mutations on genes such as frameshift & nonsense
- severity depends on where the absence is located

Alpha Thalassemia genotypes

• Normal	αα/αα
• Silent carrier	-α/αα
• Minor (trait)	-α/-α (trans)
• Hb H disease	--/αα (cis)
• Bart's hydrops fetalis	--/--

Erythropoietin increases, and can sometimes lead to extramedullary

Classification of Beta Thalassemia

- Increasing Severity ↓
- ① Beta thalassemia minor heterozygous disorder resulting in mild hypochromic, microcytic hemolytic anemia β/β^+ or β/β^0
 - ② Beta thalassemia intermedia severity lies between the minor β/β^+ and the major β^0/β^0 or β^+/ β^0 ← worse →
 - ③ Beta thalassemia major homozygous disorder resulting in severe transfusion-dependent hemolytic anemia β^0/β^0

β Thalassemia Clinical

Beta thalassemia minor

- minor point mutation (regulatory control regions) β/β^+ or β/β^0
- minimal anemia - no treatment indicated

Beta thalassemia intermedia

- homozygous minor point mutation β^+/ β^+ or more severe heterozygous β^+/ β^0
- most often do not require chronic transfusions

Beta thalassemia major - Cooley's Anemia

- Severe gene mutations (nonsense or frameshift)
- Requires intensive treatment
- Reduced or non-existent production of β globin
 - ↳ Leads to poor oxygen carrying capacity of RBC
 - ↳ Leads to failure to thrive & poor brain development
 - ↳ Increased alpha globin production & precipitation
 - ↳ RBC precursors are destroyed within the bone marrow

- Increased splenic destruction of dysfunctional RBCs
 - ↳ anemia - jaundice - splenomegaly

Hyperplastic Bone marrow

- ↳ Ineffective erythropoiesis - RBC precursors destroyed
 - ↳ Chipmunk face - poor bone growth - frontal bossing - Bone pain
- ↳ Increase in extramedullary erythropoiesis

- Iron overload - increased absorption & transfusions

- ↳ Endocrine disorders - Cardiomyopathy - Liver failure

- α chains accumulations & deposition are more toxic than β chains, therefore β Thalassemia major is more dangerous than all alpha thalassemia, but for the 4th type "Bart's hydrops fetalis"

Beta thalassemia major - Complications & emergencies

- Sepsis - Encapsulated organisms present in blood
- Cardiomyopathy
- Endocrinopathies

Beta thalassemia major - Lab findings

- Hypochromic, microcytic anemia
- Reticulocytosis (premature RBC)
- Hyperbilirubinemia
- Hemoglobin electrophoresis shows
 - ↳ increased Hgb A₂ "Delta globin production"
 - ↳ increased Hgb F "Gamma globin production"

Beta thalassemia major - treatment

① Splenectomy - indications

② Chronic Transfusion Therapy

- maximizes growth and development
- suppresses the patient's own ineffective erythropoiesis & excessive dietary iron absorption
- monthly plasma RBC transfusions

③ Chelation Therapy

- Binds free iron & reduces hemosiderin deposits

On The Horizon

- Oral Chelation Agents (deferiprone and deferasirax)
- Pharmacologically upregulating gamma globin synthesis, increasing Hgb F (hydroxy urea)
 - F hgb Carries O₂ better than Hgb A₂
 - Will help bind α globins and decrease precipitate
- Bone Marrow transplant
- Gene Therapy
 - Inserting healthy β genes into stem cells and transplanting

heterozygous individuals for a recessive gene are normal in phenotype

① Cystic Fibrosis

- The affected individual secretes thick mucus & clogs passages of lungs - pancreas - kidneys - liver - intestine
- A lung clogged by mucus won't be able to be cleared easily therefore is highly susceptible to infection by bacteria such as
Ex) *Staphylococcus aureus* & *Pseudomonas aeruginosa*
- The affected individual also has defective or absent chloride pump channels which regulate Cl^- across the cell membrane
- The sweat glands of a CF patient are abnormal, which leads to high levels of chloride in sweat
 - This chloride accumulates in cells, which will cause the cells to pick up water by osmosis from the mucus covering the cell
- Caused by mutation in CFTR gene $\Delta F508$

② Tay - Sachs disease "lysosomal storage disease" / These names will come back in the exam

- High incident among American & European Jews
- Tay-Sachs homozygotes appear normal at birth, but the symptoms appear around 8 months, then dies within a few years
 - These symptoms are seizure - deafness - blindness - degeneration of motor & mental performance
- Homozygous individuals lack the enzyme *Hexosaminidase A* which is required to break down *Gm₂ ganglioside* → Occurs in lysosomes of brain cells
- As a result the lysosome fill with gangliosides swell & eventually burst releasing oxidative enzymes that will kill the brain cells

③ Albinism "oculocutaneous albinism & ocular albinism"

- Lacks pigmentation of the skin - hair - eyes
- This is due to the inability to make brown pigment melanin
- Melanin is produced from tyrosine in the cell through enzymatic pathways
- Most albinos lack one of the enzymes required to make melanin
- Albinos display
 - ① Nystagmus, rapid uncontrolled eye movement
 - ② Strabismus, deviation of the eye from its normal axis
 - ③ Reduced visual activity, Photophobia & red eyes
- Two albinos can have a normal child only if the parents have different mutations aka lacking different enzymes
Ex) $X_0X - XX_0$ can have an XX because X_0 is recessive and not dominant

④ Phenyl ketonuria "PKU" "inborn errors of metabolism"

- PKU individuals lack the enzyme phenylalanine hydroxylase that converts phenylalanine to tyrosine
- Due to the lack of this enzyme leads to the abnormal breakdown of phenylalanine to phenyl pyruvate which will accumulate in blood-stream & urine
- Accumulation of phenyl pyruvate will lead to mental retardation
- Infants appear normal at birth, but the symptoms start to show after a few months and without treatment mental retardation will result
- Patients display
 - ① slow growth
 - ② musty odor
 - ③ fair hair & skin
- affected individuals rarely live past 30 years
- if identified at birth, they can be nutritionally treated by being put on a strict diet containing low phenylalanine
 - ↳ contains just enough for the body

Autosomal dominant inheritance

① Huntington disease "late onset disease"

- Is a degenerative disease of the nervous system, caused by a lethal dominant allele & symptoms appear around 30-40
- Degeneration of brain cells will lead to muscle spasm - loss of motor control (chorea) - dementia - affective disorders
 - ↳ Ex) Depression & Bipolar disorder
- No effective treatment, but drugs like benzodiazepines can be used to help control the choreic movements

② Neurofibromatosis "type 1 - NF1"

- Individuals will have
 - ① Café-au-lait which are hyper pigmented skin patches
 - ② Lisch nodules which is benign growth of iris
 - ③ Neurofibromas which are non-malignant peripheral nerve tumors → Benign tumor of the optic nerve

heterozygous individuals have no obvious phenotype (they are healthy) - carrier passing a faulty recessive allele - 2 faulty alleles in uterus & eggs

- Homozygous for this dominant allele is lethal & will result in spontaneous abortion
- Even though dwarfs are less likely to have children than other people, there is a high rate of mutation in the gene involved
- Cause is mutation in fibroblast growth factor receptor 3 (FGFR3) gene, which controls bone growth

Newborn Screening

- Biochemical tests like screening for DMD by measuring creatinekinase levels in new borns
- Newborn screening is effective to diagnose treatable disorders such as PKU - sickle cell - hypothyroidism - galactosemia

Direct mutation analysis

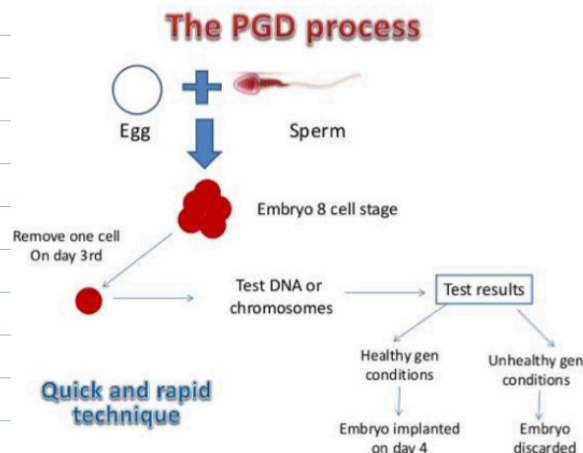
- Sometimes the mutation causing a disease happens to alter a recognition sequence for a restriction enzyme. In this case the mutation itself creates a restriction site polymorphism that can be detected after digestion with this enzyme.
- An example is sickle cell disease mutation, which alters an Mst II recognition site in the B-globin gene.

Preimplantation genetic diagnosis

- Embryo screening "PGD" is procedures performed on embryos prior to implantation
- This allows people who carry a genetic defect to avoid producing a child with a disorder

The procedure

- The procedure begins with in vitro fertilization (IVF) in which hormones (gonadotrophin-releasing hormone and recombinant follicle stimulating hormone) are used to induce ovulation.
- The ovulated eggs are surgically removed from the surface of ovary, placed in a laboratory dish, and fertilized with sperms.
- The zygotes are allowed to divide several times until they reach the 8- or 16-cell stage, at this stage one cell is removed from each embryo and later tested for the genetic abnormalities
- RT-PCR used to amplifying a minute quantities of RNA transcripts of a single cell) which make it possible to assay for the presence of specific genetic change by(DNA chips, FISH test).
- A healthy desired embryo is selected and implanted in the woman's uterus. with a thin long ~ 50 cm tube



Intracytoplasmic sperm injection

